

Work Instruction: Migrate APIM to New Subnet (Portal)

[↑ Back to README](#)

This guide provides step-by-step portal instructions with screenshots for migrating Azure API Management to a new larger subnet.

Prerequisites

Before starting, ensure: - New subnet is created in the **same VNet** - NSG with required APIM rules is attached to the new subnet - Service endpoints are enabled on the new subnet - APIM backup is completed (see [Backup Configuration](#))

Step 1: Navigate to APIM Virtual Network Settings

1. Open the Azure Portal
2. Navigate to your API Management instance
3. In the left menu, under **Deployment + infrastructure**, click **Network**
4. You will see the current VNet and subnet configuration

[image]

Current state: APIM is on `vnet-apim-01 / snet-small` (Internal mode, Southeast Asia)

Step 2: Select the New Subnet

1. Click on the **Southeast Asia** row to edit the network configuration
2. In the right panel, change the **Subnet** dropdown to the new larger subnet (`snet-large`)
3. Leave **Virtual network** as the same VNet (`vnet-apim-01`)

[image]

1. Click **Apply**
-

Step 3: Verify the New Configuration

After applying, the portal shows the updated configuration with a warning:

⚠ "To ensure your service can successfully update, please verify the new virtual network configuration before saving. Failure to do so may result in service disruptions."

Click the **Verify** button to run the VNet Verifier.

[image]

Step 4: Review VNet Verifier Results

The VNet Verifier checks your new subnet for APIM compatibility. If there are issues, you will see **Critical** or **Warning** status:

[image]

Health Status Summary showing issues:

Check	Status	Meaning
Vnet Routing Status	✓ Healthy	Route tables are correct
NSG Security Rules	✗ Critical	Missing required NSG rules
Service Endpoints	✗ Critical	Service endpoints not enabled
DNS Server	✓ Healthy	DNS resolution working
Network Connectivity	✓ Healthy	Can reach Azure services
NatGateway Status	✓ Healthy	NAT gateway configured correctly

💡 **Tip:** If you have already fixed a configuration issue but the portal still shows the old result, click on the **Time range** dropdown and re-select the time range to refresh the verifier results.

Step 5: Fix Issue — NSG Not Associated

If the new subnet doesn't have an NSG attached, you'll see this error:

[image]

✗ "Subnet doesn't have a Network Security Group associated. Without a properly configured NSG, all inbound traffic will be blocked."

Fix:

```
az network vnet subnet update \  
  --resource-group $RG \  
  --vnet-name $VNET_NAME \  
  --name $SUBNET_LARGE \  
  --network-security-group $NSG_NAME
```

Step 6: Fix Issue — Service Endpoints Not Enabled

The verifier shows which service endpoints are missing:

[image]

Required Service Endpoints:

Service	Status	Required
Azure Event Hubs	⚠ Not Enabled	If logging to Event Hub
Azure Key Vault	✗ Not Enabled	Yes
Azure Service Bus	⚠ Not Enabled	If using Service Bus
Azure SQL Database	✗ Not Enabled	Yes
Azure Storage	✗ Not Enabled	Yes
Azure Active Directory	⚠ Not Enabled	Recommended

Fix — Enable all required service endpoints:

```
az network vnet subnet update \  
  --resource-group $RG \  
  --vnet-name $VNET_NAME \  
  --name $SUBNET_LARGE \  
  --service-endpoints Microsoft.Storage Microsoft.Sql Microsoft.KeyVault \  
  Microsoft.EventHub Microsoft.ServiceBus Microsoft.AzureActiveDirectory
```

Step 7: Fix Issue — NSG Security Rules

The verifier may show failed security rule requirements:

[image]

Failed Security Rule Requirements:

Severity	Direction	Destination	Port(s)	Purpose
⚠ Warning	Outbound	EventHub	5671, 5672, 443	Log to Event Hub policy and monitoring agent
⚠ Warning	Outbound	AzureCloud	443, 12000	Health and Monitoring Extension
✖ Critical	Outbound	Internet	80	HTTPS certificate validation (CRL, OCSP, AIA)
⚠ Warning	Outbound	Internet	Any	Rule 999 Deny_All_Internet_Outbound is blocking

⚠ **Note:** All Allow rules must have a priority **lower than 999** to be evaluated before the `Deny_All_Internet_Outbound` rule. Refer to the NSG rules tables below for the complete list.

NSG Rules Reference by APIM VNet Mode

Depending on your APIM VNet mode, refer to the appropriate Microsoft documentation for the full list of required NSG rules:

VNet Mode	Documentation
External	Configure NSG rules — External VNet
Internal	Configure NSG rules — Internal VNet

Required NSG Rules — Internal APIM (stv2)

Direction	Priority	Source	Destination	Port(s)	Protocol	Purpose
Inbound	100	ApiManagement	VirtualNetwork	3443	TCP	Manage endpoint
Inbound	110	AzureLoadBalancer	VirtualNetwork	6390	TCP	Azure Infrastructure Load B

Inbound	120	AzureTrafficManager	VirtualNetwork	443	TCP	Azure Traffic Manager (routing region)
Inbound	130	VirtualNetwork	VirtualNetwork	443	TCP	Client communication to API Management
Inbound	140	VirtualNetwork	VirtualNetwork	3443	TCP	Access Azure Firewall diagnostic extension
Outbound	100	VirtualNetwork	Storage	443	TCP	Depend on Azure Storage
Outbound	110	VirtualNetwork	Sql	1433	TCP	Access Azure SQL endpoint
Outbound	120	VirtualNetwork	AzureKeyVault	443	TCP	Access Azure Key Vault
Outbound	130	VirtualNetwork	AzureMonitor	1886, 443	TCP	Publish diagnostic Logs and Metrics
Outbound	140	VirtualNetwork	AzureActiveDirectory	443	TCP	Azure AD authentication
Outbound	150	VirtualNetwork	EventHub	5671, 5672, 443	TCP	Log to Event Hub port
Outbound	160	VirtualNetwork	AzureCloud	443, 12000	TCP	Health Monitor Extension
Outbound	170	VirtualNetwork	Internet	80	TCP	Certificate validation (CRL, CAA, AIA)
Outbound	180	VirtualNetwork	Internet	25, 587, 25028	TCP	SMTP for sending emails

Outbound	999	VirtualNetwork	Internet	*	Any	Deny All (security policy)
----------	-----	----------------	----------	---	-----	--------------------------------------

Required NSG Rules — External APIM (stv2)

Direction	Priority	Source	Destination	Port(s)	Protocol	Purpose
Inbound	100	Internet	VirtualNetwork	80, 443	TCP	Client communication to APIM
Inbound	110	ApiManagement	VirtualNetwork	3443	TCP	Management endpoint
Inbound	120	AzureLoadBalancer	VirtualNetwork	6390	TCP	Azure Infrastructure Load Balancer
Inbound	130	AzureTrafficManager	VirtualNetwork	443	TCP	Azure Traffic Manager routing region)
Outbound	100	VirtualNetwork	Storage	443	TCP	Depend on Azure Storage
Outbound	110	VirtualNetwork	Sql	1433	TCP	Access Azure SQL endpoint
Outbound	120	VirtualNetwork	AzureKeyVault	443	TCP	Access Azure Key Vault
Outbound	130	VirtualNetwork	AzureMonitor	1886, 443	TCP	Publish diagnostic Logs and Metrics
Outbound	140	VirtualNetwork	AzureActiveDirectory	443	TCP	Azure AD authentication
Outbound	150	VirtualNetwork	EventHub	5671, 5672, 443	TCP	Log to Event Hub port

Outbound	160	VirtualNetwork	AzureCloud	443, 12000	TCP	Health Monitor Extensi
Outbound	170	VirtualNetwork	Internet	80	TCP	Certific validati (CRL, C AIA)
Outbound	180	VirtualNetwork	Internet	25, 587, 25028	TCP	SMTP I for send emails
Outbound	999	VirtualNetwork	Internet	*	Any	Deny A (securit policy)

Key difference: External mode requires **Inbound Internet on ports 80, 443** for public client access, while Internal mode only allows inbound from **VirtualNetwork**.



Step 8: Confirm Service Endpoints Are Added

After applying the fixes, verify the service endpoints are enabled on the subnet:

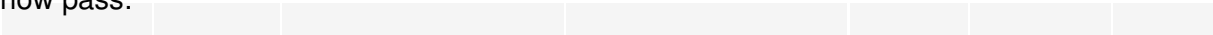
[image]

Required service endpoints configured: - ✓ Microsoft.EventHub - ✓ Microsoft.KeyVault - ✓ Microsoft.ServiceBus - ✓ Microsoft.Sql - ✓ Microsoft.Storage - ✓ Microsoft.AzureActiveDirectory



Step 9: Re-run VNet Verifier — All Passed

After fixing NSG rules and service endpoints, click **Verify** again. All checks should now pass:



[image]

Health Status Summary:

Check	Status
Vnet Routing Status	✓ Healthy

NSG Security Rules	✓ Healthy
Service Endpoints	✓ Healthy
DNS Server	✓ Healthy
Network Connectivity	✓ Healthy
NatGateway Status	✓ Healthy

✓ "NSG Security rules fulfill all of the recommended requirements." ✓
 "Service endpoints are enabled for all recommended services." ✓ "Network connectivity check with all of the required dependencies succeeded."

Step 10: Save the Configuration

Once all verifications pass, click **Save** to start the subnet migration.

[image]

⚠ **This triggers the migration. APIM will experience downtime (15-45 minutes).**

Step 11: Migration In Progress

After saving, the portal shows a notification that the service is being updated:

[image]

i **"Service is being updated - this operation might take up to 160 minutes. During the service update, certain features will be temporarily unavailable."**

The APIM network page shows the new subnet (`snet-large`) is now configured.

Step 12: Verify Migration Complete

Once the operation completes, verify the APIM overview shows the new VIP addresses on the larger subnet.

Before Migration (on `/28` subnet)

[image]

Field	Value
Status	Online
Tier	Premium
Public VIP	x.x.x.x
Private VIP	10.2.0.4
Subnet	snet-small (/28)

After Migration (on /25 subnet)

[image]

Field	Value
Status	Online
Tier	Premium
Public VIP	x.x.x.x (changed)
Private VIP	10.2.1.4 (changed)
Subnet	snet-large (/25)

⚠ **Important: Both public and private IP addresses change after subnet migration.** - Update **Private DNS zone** to point to the new private IP - Update **Application Gateway** backend pool with new private IP - Update **Firewall rules** to allow traffic to/from new subnet range - Notify downstream consumers if they reference the public VIP directly

Step 13: Verify Scaling is Now Available

After migration to the larger subnet, confirm that APIM can now scale beyond the previous limit. Navigate to **Monitoring** → **Scale out (auto-scale)**:

[image]

✓ **The instance count can now be increased to 10 (or more)** — this was previously blocked by the /28 subnet running out of IP addresses. The larger

/25 subnet provides 123 usable IPs, sufficient for scaling well beyond 10 units.

Post-Migration Actions

After successful migration, complete these steps:

1. **Update DNS:** Point Private DNS zone to the new private IP (10.2.0.4)
2. **Update firewall rules:** Allow traffic to/from the new subnet range
3. **Update Application Gateway:** If applicable, update backend pool with new IP
4. **Scale APIM:** Now safe to scale to desired unit count
5. **Test APIs:** Verify API calls work from within the VNet

```
# Verify final state
az apim show --resource-group $RG --name $APIM_NAME \
  --query "{name:name, status:provisioningState, privateIPs:privateIPAddresses}" \
  -o json
```

[↑ Back to README](#)